

The v3 Controller/Estimator Campaign on the Mecanum Digital-Twin Plant

What was run, why, and how to read it — explainer companion to `RESULTS_v3.md`

Mecanum-PINN dynamics notes
KUKA youBot four-Mecanum platform, IMECE 2026 track
code: `hybrid_ctrl_v2/` (Julia closed-loop stack)

August 20, 2026

Abstract

This note explains the v3 controller/estimator simulation campaign end to end: the question it answers (§1), the simulated plant, sensors and estimator (§2), the feasibility screening that defines which trajectories are fair to ask for (§3), the scalar objective every run minimizes (§4), the five-stage protocol (§5), the headline findings (§6), and the reading rules that keep the numbers honest (§7). It is the narrative companion to `hybrid_ctrl_v2/RESULTS_v3.md`, which remains the document of record for the numbers themselves.

1 The question

The project needs a defensible answer to: *which control law should the omnidirectional platform fly, and how much of its performance survives real state estimation?* Three candidate laws are compared, chosen so the comparison is a two-step ablation:

Controller	Structure	What it isolates
PID-FB	cascaded IMC-reparameterized PID, 3 tuned λ_{inner} per axis, rest derived	industrial baseline
PID-CT	same cascade + computed-torque feedforward (matched to ASMC’s equivalent control)	FB→CT: value of the <i>model</i>
ASMC v2	adaptive-gain supertwisting SMC, friction-circle gain ceiling recomputed per tick	CT→ASMC: value of <i>adaptation</i>

All three are tuned by the same black-box optimizer against the same scalar objective on the same trajectory tiers, then evaluated through the same frozen state estimator. Anything that makes one controller’s objective differ from another’s is treated as a bug, not a detail. MPC was part of the original plan and is deferred out of this iteration (diverges on four velocity-demanding trajectories; its `use_ltv=false-is-better` anomaly remains unresolved).

2 The system under test

Plant. The 30-state voltage-input plant (`hybrid_ctrl/plant.jl`): slip-spin-LuGre contact physics (per-wheel bristle states, passive roller spin) driven through a DC-motor model with

back-EMF,

$$\tau = G\eta(K_t i - \tau_f), \quad i = (V - K_b G\omega)/R_a, \quad |V| \leq 24 \text{ V}, \quad (1)$$

with a 200 V/s slew clamp in the mixer. Actuator authority falls linearly with wheel speed and vanishes at the no-load speed $V_{max}/(K_b G) = 27.73 \text{ rad/s}$ — the hard feasibility wall of §3. (Derived in docs/Mecanum_Analytical_Limits_AxisVel_AccelEnvelope_BackEMF.tex.)

Sensors (:realistic suite). IMU and wheel encoders at 1000 Hz; optical flow at 100 Hz with 2% dropout; 100 Hz pose fixes with per-run biases and 0.5% outliers — industrial white noise plus deliberately uncalibrated consumer-grade biases. Neither the estimator nor (in the deployable configuration) the controllers ever see the true state.

Estimator. ESKFestimatorV3: a 13-state error-state Kalman filter with a yaw-acceleration random-walk state (a type-2 tracker, added to remove a structural yaw-rate lag). It is tuned once and **frozen** before any controller comparison — the *estimator-first methodology* — so a control-law difference can never be an observer difference in disguise.

Three feedback regimes — not two. This distinction runs through the whole campaign; getting it wrong inverts conclusions:

1. **clean oracle** — perfect state feedback; an upper bound.
2. **noisy oracle** — true state *plus raw, unfiltered sensor noise* written into the controller’s state estimate. There is *no estimator* in this path; it is a pessimistic no-filtering bound, not a deployment model.
3. **ESKF** — the deployable path: **:realistic** sensors through the frozen filter.

The ESKF beats the noisy oracle *by construction* (a tuned filter beats raw noise), so “ESKF vs noisy oracle” is **not** the cost of state estimation — that cost is ESKF vs *clean* oracle.

3 Feasibility first: the v3 trajectory tiers

The predecessor tier (**train12**) contained trajectories the hardware cannot fly, and the tuner was spending most of its budget on them: **spiral_orbit_stress** alone carried 72–81% of the training tracking term while sitting at 85.4% of the motor’s no-load speed. The v3 tiers (**train14_v3**, **test_v3**) screen every candidate reference over its own full duration against two walls:

1. **Motor / back-EMF (hard):** max per-wheel speed $(V_x \pm V_y \pm (l+h)\dot{\psi})/R$ against the 27.73 rad/s no-load speed; past $\sim 85\%$ of no-load no gain law can arrest error growth. v3 targets $\leq 70\%$ for margin.
2. **Friction circle (soft):** per-wheel contact utilisation. Brief excursions are momentary slip the LuGre plant absorbs — measured to be harmless — so this wall informs but does not veto.

The point of the rebuild: a controller comparison is only meaningful where the hardware can actually go.

4 The objective — one scalar, three terms

$$\text{score} = \text{tracking}_{v3} + 0.05(\text{ce}/V_{MAX}) + 0.36(\text{chatter}/\text{CHATTER}_{REF}). \quad (2)$$

tracking_{v3}: six tolerance-normalised terms (position and heading $\times$ terminal / peak / typical), divided by a per-trajectory extent scaler k_{traj} . The v2 metric scored four samples out of 12k–60k per run (terminal and argmax only) and was blind to mid-path oscillation; the time-normalised integral (“typical”) term is the fix, with tolerances calibrated from measured achieved ratios. A term of 1.0 means “exactly at target”. **ce**: mean summed wheel-voltage magnitude (energy proxy). **chatter**: mean per-tick total variation of the wheel-voltage command against $\text{CHATTER}_{REF} = 0.8 \text{ V/ms}$, the mixer’s own slew clamp. This term exists because an entire arc of v2 tuning kept railing on chatter the objective could not price (chatter had been normalised by V_{MAX} — a voltage dividing a slew rate, $\sim 30\text{--}100\times$ too weak). $\lambda_{chatter} = 0.36$ is *balance-preserving, not re-derived*: it reproduces the accepted v2 tracking share (29.9%) under the rescaled v3 metric, and the objective code refuses $\lambda_{chatter} > 1.44$ under `--metric v3` so a carried-over v2 value cannot pass silently.

The estimator has its own objective. A *replay* score (estimator vs ground truth on logged trajectories, no controller in the loop), same tolerance-normalised shape, with velocity error *during slip* priced twice (`lam_slip=1.0`): the point of an estimator on this platform is that its model mismatch *is* slip. It converged to $\sim 71\%$ a velocity-estimation objective, matching what the controllers’ inner loops consume.

5 The protocol — five stages, in order

1. **Estimator tuning (first, then frozen).** 5 seeds over a 7-dim log-scaled gain space, replay objective, `:realistic` sensors. All seeds converge to the same estimator within 0.23% on a common noise realisation; seed 4 adopted and frozen for the whole campaign — including tiers it was not tuned on.
2. **Feasibility screening** of trajectory tiers (§3).
3. **Clean controller tuning** — 5 seeds each, cold start, oracle-clean feedback, `train14_v3`, convergence by plateau detection (the v1 tuner was caught terminating mid-descent, which invalidated its scatter claims).
4. **Noisy controller tuning** — 4 seeds each, warm-started from the same run’s clean optimum, feedback = noisy oracle. Run for all three controllers or none, so a noisy-tuned ASMC is never quoted against clean-tuned PIDs.
5. **Evaluation under three feedback regimes** — the six converged configs (3 controllers $\times$ clean/noisy-tuned) on fresh sensor seeds 101–105: oracle-clean, oracle-noisy, and through the frozen ESKF (420 runs: 6×14 trajectories $\times$ 5 seeds, zero non-converged). The held-out `test_v3` leg (generalisation of the final estimator+controller stack) completes the campaign: 288 oracle sims + 240 ESKF runs — §6 and `RESULTS_v3.md` §9.

Throughout: one OS process per seed (the pipeline has global mutable state), BLAS pinned to one thread for determinism, `stop_reason` recorded per run, and outputs under `runs_*/` never hand-edited.

6 Headline results

Full tables and caveats in `RESULTS_v3.md`; the structural findings:

PID-CT wins every regime. Clean oracle 0.14340 vs ASMC 0.15318 (no seed overlap; ordering $\text{PID-CT} < \text{PID-FB} < \text{ASMC}$); noisy oracle; and frozen-ESKF closed loop 0.34127, paired 69/70

and 70/70 against ASMC and PID-FB.

Noisy tuning does not transfer to deployment. It helps against the noisy oracle it was tuned on (ASMC 5/5, CT 5/5 paired) but the advantage vanishes through the real ESKF, and it *damaged* ASMC’s heading $\sim 2\times$ everywhere. Deploy clean-tuned configs.

Real state estimation costs +0.20 of score, uniformly. Against clean oracle, essentially identical across all six configs (42.9–45.1% of the clean $\rightarrow$ unfiltered gap recovered): the estimator’s contribution is controller-independent — exactly what makes a frozen observer a fair basis for comparison.

The error budget is a three-stage story. Sensor injection (identical across configs by construction) $\rightarrow$ ESKF survival (74–79% of injected velocity error removed; pose removal flat at $\sim 67.6\%$ on every trajectory) $\rightarrow$ controller tracking, where essentially *all* inter-controller spread lives. The estimator is common mode; only the tracking stage discriminates control laws.

CT’s one weakness is heading under sustained yaw. Best heading on 9 of 14 trajectories (~ 1.1 mrad), collapsing to 29.8/37.6 mrad on exactly the two sustained-yaw stress trajectories — the same two where its yaw integral clamp fires on 17–20% of ticks (0.0% elsewhere). Three independent measurements agree; whether the clamp binding *harms* (vs merely binds) is not yet established.

Much of the tier is at the estimator’s noise floor. On 7 of 14 trajectories, tracking error ($\sim 2\text{--}3$ mm) sits at or below the estimator’s own pose accuracy (~ 3.2 mm): further controller improvement there is unmeasurable with this sensor suite, so the comparison is decided by the handful of trajectories where tracking rises clear of the floor.

Heading is the ESKF’s best channel; yaw rate is its worst. Heading sees $6.9\times$ error removal once an 11 ms startup transient on non-zero-initial-heading references is accounted for; yaw rate sees only $\sim 52\%$ removal — the gyro is its only source, with no absolute fix to lean on.

Generalisation (test_v3). The held-out tier answers whether any of the above transfers to trajectories the tuning never saw. Three cautions and one headline:

- **Anchor caveat.** `coupled_vomega_anchor` is combo-identical to the training tier’s `coupled_vomega_stress` — a deliberate cross-tier consistency check, which passes bit-exactly (± 0.000000) — and must be excluded from every generalisation claim. It is the worst-tracked entry in either tier, so leaving it in would dominate any mean. The tier therefore yields **7 genuinely held-out trajectories**, 3 on profiles never seen in tuning.
- **PID-CT generalises as the winner, unambiguously.** Clean-tuned PID-CT beats ASMC and PID-FB **35/35 paired** each on the held-out set and is first on every individual held-out trajectory — the campaign’s most robust result.
- **The ASMC-vs-PID-FB ordering does not generalise.** Their training-tier gap was carried by a single trajectory (`coupled_vomega_stress`, 66% of ASMC’s gap); excluding it, they are statistically indistinguishable (23/35 and 8/35). The defensible claim: **ASMC and PID-FB are equivalent, and PID-CT beats both.**

- **The estimator is tier-invariant.** Every error-budget channel lands within ~ 3 percentage points of its training value, and on all 7 held-out trajectories every controller tracks to 1.6–4.2 mm / 1.0–2.5 mrad. The heading collapses occur only on the anchor — a *training* trajectory — consistent with CT’s yaw-clamp weakness being a sustained-yaw property, not a generalisation failure.

7 Reading rules

1. **Never mix tuning-stage scores with eval scores** — different noise realisations, overlapping ranges.
2. **Pair, don’t average, across realisations** — realisation variance ($\sim 6\text{--}7\%$ eval, $\sim 16\%$ tuning) swamps controller differences ($\sim 2\text{--}4\%$). Every ordering claim is a paired 5/5 or 4/4, never a difference of marginal means.
3. **Best-seed selection under noise is confounded** — seed 4 is both the best seed and the easiest realisation for all three controllers; applied identically, so cross-controller comparison holds.
4. **Absolute values, never % degradation** — a better clean tracker reads a worse % for the same absolute result.
5. **Three feedback regimes, not interchangeable** — §2.

8 Reproduction map

Piece	Where
plant (untouchable v1)	hybrid_ctrl/plant.jl, scheduler.jl, mixer.jl
v2 controllers + search spaces	hybrid_ctrl_v2/controllers_v2.jl, tune_controller_v2.jl
objective + trajectory tiers	controller_tuning/stage_objective.jl, trajsets.jl
tuning drivers	run_stage_asmc_v3.bat, run_stage_pid_v3.bat (noisy warm-starts)
estimator tuning	estimator_tuning/ (objective_v2, param_space_v4, replay)
frozen estimator config	runs_estimator_v4_mu0p5_train12/seed4/best_config.json
converged controller configs	runs_asmc_v3/, runs_pid_v3/
closed-loop ESKF eval	eval_v3_eskf.jl, data runs_eskf_v3_train14/
diagnostics	diag_v3_pid_imax_binding.jl, diag_v3_error_budget.jl
provenance / verification	metric_v3_{calib,verify}.jl, scalars_v3.jl, lambda_v3_verify.jl
feasibility theory	docs/Mecanum_Analytical_Limits_AxisVel_AccelEnvelope_BackEMF.tex
s=1.0 ablation	run_stage_{asmc,pid}_v3_s100.bat, eval_v3_s100.jl, RESULTS_v3_s100.md
significance tests	stats_significance_s100.py (+ .bat)

9 Deliberately not done (yet)

- **MPC** — deferred (§1).
- **test_v3 generalisation** — **done** (§6): PID-CT generalises 35/35; ASMC/PID-FB tie. Caveat: the tier yields 7 held-out trajectories, not 8 (anchor is combo-identical to a training entry).
- **CT yaw integral budget sensitivity** — the clamp is measured to *bind*; that it *harms* is unmeasured.
- **ESKF yaw-rate re-weighting** — interacts with the CT heading finding; the obvious next experiment.

- $\hat{\psi}$ **initialisation from the first pose fix** — removes the 11 ms startup transient that makes heading statistics incomparable across mixed-initial-heading tiers.

10 Follow-up ablation: the friction-circle safety margin ($s = 1.0$)

Every result so far carries a 0.9 factor of safety on the *available* friction circle (the same circle the feasibility envelope of `Mecanum_Analytical_Limits_AxisVel_AccelEnvelope_BackEMF.tex` derives; §A). The supervisor’s question: is that margin hiding performance? The factor enters exactly two controller gates — ASMC’s `kmax_schedule` gain ceiling and PID-CT’s `vcmd_limits` command gate γ (the V_{ff} ceiling) — so the ablation re-runs the *entire* v3 chain for ASMC and PID-CT with `safety_margin = 1.0`, single-variable, same seeds and optimizer budgets. PID-FB is out of scope (no feedforward). Verified gate deltas: K_{max} free-circle ceiling $\times 1.1111$ ($= 1/0.9$ exactly), loaded circle $\times 1.1452$ (nonlinear), γ $0.4388 \rightarrow 0.4876$; the default path is byte-identical to the archived campaign, outputs auto-suffixed `_s100`, and the margin is recorded in each config and read back at eval time, so gates can never be mixed across campaigns. Full tables: `hybrid_ctrl_v2/RESULTS_v3_s100.md`.

Clean tuning: a real, consistent gain. All ten $s=1.0$ seeds beat their family’s archived best — ASMC 0.15197 vs 0.15318 (-0.8%), PID-CT 0.14175 vs 0.14340 (-1.2%). This is the regime where the gates bind.

But it does not survive noise. The noisy-tuning objective is flat to slightly worse (paired means $+0.0039$ ASMC, $+0.0011$ PID-CT against a realisation std of ≈ 0.037); oracle noisy-means move ≤ 0.003 on `train14_v3`. Through the frozen ESKF, only one config moves materially: ASMC clean-tuned improves on both tiers ($-0.0034 / -0.0061$) with slightly lower chatter everywhere; the rest are unchanged. The ordering **PID-CT < ASMC is untouched at every stage and tier**, and the noisy-tuning findings replicate (helps ASMC > PID-CT, 5/5 seeds, smaller at the relaxed margin).

The tuner never rails the new ceilings. ASMC’s clean optimum moves to *lower* surface slopes (λ_x^{max} $5.68 \rightarrow 3.48$); PID-CT’s $\lambda_{inner,x}$ doubles ($0.098 \rightarrow 0.256$). The margin was shaping the achievable gain landscape, not truncating an optimum sitting against it.

Significance. Paired on (trajectory $\times$ noise seed), run-level ($n = 70/40$) and traj-level (seeds averaged, $n = 14/8$), paired $t +$ Wilcoxon (`stats_significance_s100.py`):

- **Controller gap:** PID-CT – ASMC is significant at both margins, both tiers, both tuning regimes — worst case $p = 9.7 \times 10^{-4}$, Cohen’s d_z 0.56–0.87. The ordering is a property of the platform, not of the margin.
- **Margin effect:** of eight tests, only **ASMC clean-tuned** is significant — worst $p = 0.015$, 21/22 trajectories better, `test_v3` $d_z = -1.46$, Holm-surviving. PID-CT ct’s `test_v3` move ($p_t = 0.059$) is a trend, not a finding.

Scope: inference covers the sensor-noise distribution and this fixed trajectory set (design points, not random draws); and even the significant effect is $\sim 1\%$ of score — real, consistent, small.

Verdict. The 0.9 factor of safety costs $\sim 1\%$ of clean-regime tracking-objective headroom and buys nothing statistically robust under noise beyond one small ASMC-ct improvement. Keeping it or not is now a quantified design call.

A Plant model and constants

All campaign runs share the case-1 parameter set (`PlatformParams/LuGreParams + MotorParams, hybrid_ctrl/plant.jl`); estimator tuning and all evaluations at $\mu = 0.5$, $\chi = 0.005$ (dataset grid $\{0, \dots, 0.008\}$; 12 rollers per wheel). Derivations from these numbers (velocity caps, envelope, two-gate feasibility) live in `docs/Mecanum_Analytical_Limits_AxisVel_AccelEnvelope_BackEMF.tex`.

A.1 Constants

Table 1: Platform, geometry and contact (O-configuration: $\delta = (-45^\circ, +45^\circ, +45^\circ, -45^\circ)$; wheel centres $p_x = (h, h, -h, -h)$, $p_y = (l, -l, l, -l)$).

Sym	Value	Meaning	Sym	Value	Meaning
h	0.235 m	half-length	m	30 kg	body mass
l	0.15 m	half-width	m_s	35.6 kg	total mass
R	0.05 m	wheel radius	N_{tot}	349.2 N	total normal
R_a	0.0355 m	roller axle dist.	J_w	$5.87 \times 10^{-3} \text{ kg m}^2$	wheel inertia
$R - R_a$	0.0145 m	—	J_{roller}	10^{-6} kg m^2	roller inertia
μ	0.5	Coulomb coeff.	a_X	0.016 m	COM offset (long.)
$s = \mu_s/\mu_c$	1.1	stiction ratio	a_Y	-0.026 m	COM offset (lat.)
v_s	0.01 m/s	Stribeck vel.	p_1	0.11	wheel viscous
χ	0.005	contact-spot	p_2	5.78×10^{-3}	roller viscous

Table 2: DC-motor actuator (`MotorParams`) and derived landmarks.

Sym	Value	Meaning	Sym	Value	Meaning
$K_t = K_b$	0.0335	torque / back-EMF const.	V_{max}	24 V	bus voltage
G	$9405/364 \approx 25.84$	gear ratio	dV_{max}	200 V/s	mixer slew clamp
$R_a^{(m)}$	2.0 Ω	winding resistance	i_{max}	12.8 A	current limit (slack)
η	0.9	gear efficiency	τ_f	0.01 N m	shaft friction
τ_{stall}	9.12 N m	derived, (12)	ω_{nl}	27.73 rad/s	no-load (hard wall)

A.2 Plant equations (what the simulator integrates)

Per-contact slip velocities (representative roller phase $\tilde{\varphi}_i \rightarrow 0$; roller-spin rate $\dot{\gamma}_i$, wheel rate ω_i):

$$V_{px,i} = V_x - \dot{\psi} p_{y,i} - \omega_i R + \dot{\gamma}_i \sin \delta_i (R_a - R), \quad (3)$$

$$V_{py,i} = V_y + \dot{\psi} p_{x,i} + \dot{\gamma}_i \cos \delta_i (R - R_a), \quad \omega_{z,i} = \dot{\psi}. \quad (4)$$

Steady LuGre/Stribeck contact force (bounded, aligned with slip):

$$\vec{F}_i = -N_i g_t(\|V_p\|) \frac{\vec{V}_{p,i}}{\|V_p\|}, \quad g_t = \mu_c \left[1 + (s - 1) e^{-(\|V_p\|/v_s)^2} \right], \quad (5)$$

with per-contact bristle states integrated in the full plant (30 states: 3 body velocities, ψ , 4 wheel angles, 4 wheel rates, 4 roller rates, 2 world positions, 12 bristle states).

Wheel (actuated) and roller (passive) dynamics:

$$J_w \dot{\omega}_i = M_i - p_1 \omega_i - F_{x,i} R, \quad (6)$$

$$J_r \ddot{\gamma}_i = -p_2 \dot{\gamma}_i - F_{x,i} \sin \delta_i (R - R_a \cos \tilde{\varphi}_i) - F_{y,i} \cos \delta_i (R_a - R \cos \tilde{\varphi}_i) - M_{z,i} \sin \tilde{\varphi}_i \cos \delta_i. \quad (7)$$

Platform rows (COM offset retained; M^{-1} applied to the RHS):

$$\tilde{m} \dot{V}_x - m a_Y \ddot{\psi} = \sum_i F_{x,i} + m_s \dot{\psi} V_y + m a_X \dot{\psi}^2, \quad (8)$$

$$\tilde{m} \dot{V}_y + m a_X \ddot{\psi} = \sum_i F_{y,i} - m_s \dot{\psi} V_x + m a_Y \dot{\psi}^2, \quad (9)$$

$$-m a_Y \dot{V}_x + m a_X \dot{V}_y + I_s \ddot{\psi} = \sum_i (p_{x,i} F_{y,i} - (p_{y,i}) F_{x,i} + M_{z,i}) - m \dot{\psi} (a_X V_x + a_Y V_y). \quad (10)$$

Actuator (quasi-static DC motor, evaluated every RHS call):

$$i = \frac{V_{cmd} - K_b G \omega_i}{R_a^{(m)}}, \quad M_i = G \eta (K_t i - \tau_f \text{sign } \omega_i), \quad |V_{cmd}| \leq V_{max}, \quad |\dot{V}_{cmd}| \leq dV_{max}. \quad (11)$$

The torque-speed line and its landmarks:

$$\tau_{avail}(\omega) = \underbrace{G \eta \left(\frac{K_t V_{max}}{R_a^{(m)}} - \tau_f \right)}_{9.12 \text{ N m}} - \underbrace{\frac{G^2 \eta K_t K_b}{R_a^{(m)}}}_{0.337 \text{ N m s/rad}} |\omega|, \quad \omega_{nl} = \frac{V_{max}}{K_b G} = 27.73 \text{ rad/s}. \quad (12)$$

No-slip wheel-speed map (used by feasibility screening and the estimator's wheel Jacobian):

$$\omega_i = \frac{1}{R} \left[V_x + \tan \delta_i V_y + (\rho_{X,i} \tan \delta_i - \rho_{Y,i}) \dot{\psi} \right] = \frac{V_x \pm V_y \pm (l+h) \dot{\psi}}{R}. \quad (13)$$

Equivalent (augmented) 3-DOF model the controllers are designed on:

$$M_{aug} \ddot{\vec{v}} + C(\dot{\psi}) \dot{\vec{v}} + D_{eq} \vec{v} + \vec{f}_c(\vec{v}) = \vec{F}_p, \quad \vec{v} = (V_x, V_y, \dot{\psi}), \quad (14)$$

$$M_{aug} = \begin{bmatrix} \tilde{m} & 0 & -m a_Y \\ 0 & \tilde{m} & m a_X \\ -m a_Y & m a_X & I_\psi \end{bmatrix}, \quad \tilde{m} = m_s + \frac{4J_w}{R^2}, \quad I_\psi = I_s + \frac{4(l+h)^2}{R^2} J_w, \quad (15)$$

$$D_{eq} = \text{diag} \left(\frac{4p_1}{R^2}, \frac{4p_1}{R^2} + \frac{8p_2}{(R-R_a)^2}, \frac{4p_1(l+h)^2}{R^2} + \frac{8p_2 h^2}{(R-R_a)^2} \right) \approx \text{diag}(176, 396, 38). \quad (16)$$

B The frozen estimator (ESKF v3) — complete equations

State (error-state KF, ESKFestimatorV3, `hybrid_ctrl_v2/estimators_v3.jl`; heading carried as $(c, s) = (\cos \psi, \sin \psi)$, re-normalised each tick):

$$x = [V_x, V_y, \dot{\psi}, c, s, X, Y, b_x, b_y, s_x, s_y, b_g, \alpha]^\top \in \mathbb{R}^{13}, \quad (17)$$

with accelerometer biases (b_x, b_y) , slip states (s_x, s_y) , gyro bias b_g , and the 13th state α = yaw acceleration (the v3 addition: a second integrator on the yaw channel; a type-2 tracker follows a constant-acceleration ramp with zero steady-state lag, which the type-1 v2 design structurally could not).

B.1 Prediction (1 kHz, IMU acceleration as input)

$$V_x^+ = V_x + dt(a_x + \dot{\psi}V_y - b_x), \quad V_y^+ = V_y + dt(a_y - \dot{\psi}V_x - b_y), \quad (18)$$

$$\dot{\psi}^+ = \dot{\psi} + dt\alpha, \quad \alpha^+ = \alpha + w_\alpha, \quad (19)$$

$$c^+ = c - dt\dot{\psi}s, \quad s^+ = s + dt\dot{\psi}c, \quad (20)$$

$$X^+ = X + dt(V_x c - V_y s), \quad Y^+ = Y + dt(V_x s + V_y c), \quad (21)$$

$$s_x^+ = (1 - dt/\tau_{slip})s_x, \quad s_y^+ = (1 - dt/\tau_{slip})s_y, \quad (22)$$

biases (b_x, b_y, b_g) constant in prediction; $\tau_{slip} = 0.06$ s, $\sigma_{slip} = 0.014$ m/s (measured from traces). Slip states are *mean-reverting* (Ornstein–Uhlenbeck) in both the state and the error Jacobian — v1’s pure random walk has no stationary variance.

B.2 Error dynamics and process noise

$P^+ = FPF^\top + Q$, $F = I + dtA$, A ’s nonzero entries read off the prediction above, including the yaw double-integrator coupling $A_{3,13} = 1$ and the slip decay $A_{10,10} = A_{11,11} = -1/\tau_{slip}$. Process noise, all entries derived (none searched):

$$Q_{11} = Q_{22} = (dt\sigma_{acc})^2 q_{scale}, \quad q_{scale} = 1 + \alpha_{acc}\|a\| + \alpha_{yaw}|\dot{\psi}|, \quad (23)$$

$$Q_{33} = (dt\sigma_{gyro})^2 q_{scale}, \quad Q_{44} = Q_{55} = \text{pose_Qn_heading}, \quad (24)$$

$$Q_{66} = Q_{77} = \text{pose_Qn_pos}, \quad Q_{88} = Q_{99} = 10^{-10} \text{ (margin)}, \quad (25)$$

$$Q_{12,12} = \sigma_{rw}^2 dt \text{ (gyro random walk)}, \quad Q_{13,13} = q_\alpha \text{ (tuned)}. \quad (26)$$

$$Q_{10,10} = Q_{11,11} = \frac{2\sigma_{slip}^2}{\tau_{slip}} dt \times \begin{cases} g_{grip} & \text{gripping} \\ 1 & \text{slipping} \end{cases} \quad (g_{grip} = \text{grip_slip_scale}). \quad (27)$$

B.3 Wheel + gyro update (1 kHz)

Encoder wheel speeds $\tilde{\omega}$ are least-squared through the O-config Jacobian (13) — $H_\omega^\top H_\omega = \text{diag}(4/R^2, 4/R^2, 4(l+h)^2/R^2)$, so the body-velocity pseudo-measurement covariance is closed-form, exactly diagonal and isotropic — plus the gyro:

$$z = \begin{bmatrix} (H_\omega^+ \tilde{\omega})_x \\ (H_\omega^+ \tilde{\omega})_y \\ \tilde{g}_z \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 & 0 & \cdots & s_x \leftarrow x_{10} & \cdots \\ 0 & 1 & 0 & \cdots & s_y \leftarrow x_{11} & \cdots \\ 0 & 0 & 1 & \cdots & b_g \leftarrow x_{12} & \cdots \end{bmatrix}_{3 \times 13}, \quad (28)$$

$$R = \text{diag}(R_w, R_w, R_g), \quad R_w = \frac{(\sigma_\omega^2 + (sf_\omega \tilde{\omega})^2)R^2}{4}, \quad R_g = \sigma_{gyro}^2 + (sf_{gyro} \dot{\psi})^2. \quad (29)$$

While slipping (R inflated `×slip_R_inflate`; accel-bias gains zeroed), and with an NIS gate at $\chi_{3,0.99}^2 = 11.34$ that on violation boosts the wheel rows `×r_boost` and retries once. Slip detection is pinned, not tuned: $\text{slip} = \|(H_\omega^+ \tilde{\omega}, \tilde{g}_z) - (V_x, V_y, \dot{\psi})_{pred}\|$, threshold = $3\sqrt{R_w}$. Joseph-form covariance update with symmetrisation throughout.

B.4 Flow update (100 Hz) and pose fix (100 Hz)

Flow (skipped entirely on dropout or loss-of-lock — *never* a zero substitution):

$$H_{flow} = \begin{bmatrix} 1 & 0 & -r_y & 0 & \cdots & 0 \\ 0 & 1 & r_x & 0 & \cdots & 0 \end{bmatrix}_{2 \times 13}, \quad R_{flow} = (\sigma_{flow0} + \sigma_{flow}|v|)^2 + \frac{\text{quantum}^2}{12}, \quad (30)$$

lever arm $r = (0, 0)$ in this campaign; the *slip columns are zero* — wheels measure $V_x + s_x$, flow measures V_x , which is what breaks the velocity $\leftrightarrow$ slip degeneracy. NIS gate $\chi_{2,0.999}^2 = 13.82$; α informed only through covariance.

Pose fix (:docking tier): $z_4 = (x, y, \cos \psi, \sin \psi)$, H selects (X, Y, c, s) , $R = \text{diag}(\sigma_{pos}^2, \sigma_{pos}^2, \sigma_{\psi}^2, \sigma_{\psi}^2)$, NIS gate $\chi_{4,0.999}^2 = 16.27$.

Output to controllers: $\hat{x} = [V_x, V_y, \dot{\psi}, \text{atan2}(s, c), X, Y]$, $\hat{d} = [s_x, s_y, 0]$. Divergence guard: innovation NIS $> 10^4$ on 10 consecutive ticks rescales $P \times 10$.

B.5 Converged configuration (adopted seed 4)

Tuned (runs_estimator_v4_mu0p5_train12/seed4/best_config.json): $\alpha_{acc} = 0.0229$, $\alpha_{yaw} = 0.0146$, grip_slip_scale = 3.66×10^{-4} , r_boost = 61.6, slip_R_inflate = 42.2, pose_Qn_heading = 10^{-8} , $q_\alpha = 0.0666$. Pinned physical priors (“ P_0 is a statement of knowledge”): $P_0^{vel} = 10^{-4}$, $P_0^{yaw} = 10^{-3}$, $P_0^{head} = 0.5$, $P_0^{bias\,acc} = 4 \times 10^{-4}$, $P_0^{bias\,gyro} = 9 \times 10^{-6}$, $P_0^{slip} = 2 \times 10^{-4}$, $P_0^{pos} = 0.25$, $P_0^\alpha = 0.25$, pose_Qn_pos = 10^{-7} .

C Exact noise specification

Every noise figure the campaign ran under, verbatim from the implementation. *Sources:* hybrid_ctrl_v2/sensors_v2.jl (:realistic suite, Tables 3–6) and hybrid_ctrl/estimators.jl (OracleEstimator, Table 7); datasheet provenance per field is in instructions/sensors-suite-consolidation-and-physical-noise.md, Appendix A (reference parts: ADIS16470 IMU; PixArt PMW3901 / ADNS-3080 optical flow). Grade claim: *industrial white-noise levels with uncalibrated consumer bias terms*. Each modality draws from its own RNG seeded `hash((seed, modality))`, so measurement sequences are independent of callback order.

Table 3: IMU, 1000 Hz (ImuModel; :realistic sets sf_gyro).

Field	Value	Provenance note
sigma_acc	0.05 m/s ²	= 228 $\mu\text{g}/\sqrt{\text{Hz}}$; typical MEMS 60–300
sigma_gyro	0.003 rad/s	= 0.0077 $^\circ/\text{s}/\sqrt{\text{Hz}}$; ADIS16470 spec 0.008
acc_bias	0.02 m/s ²	= 2.04 mg; consumer uncalibrated (ADIS16470 in-run: 13 μg)
bias_gyro	$0.003 \cdot \mathcal{N}(0, 1)$ rad/s	turn-on bias, drawn once per run ($\sim 619^\circ/\text{hr}$ scale)
gyro_bias_rw	10^{-4} rad/s/ $\sqrt{\text{s}}$	proper Wiener increment; $\rightarrow 92^\circ/\text{hr}$ at $T=20$ s
sf_gyro	0.005	$\dot{\psi}$ -proportional white scale factor (heteroscedastic R_g)

Table 4: Wheel encoders, 1000 Hz (`EncoderModel, :realistic`).

Field	Value	Provenance note
<code>enc_cpr</code> $\times$ gear	4000×25.84	motor-shaft counts, quantized
<code>sigma_omega</code>	$0.01/(R\sqrt{1.09}) = 0.192$ rad/s	quadrature-derived speed noise
<code>sf_omega</code>	0.02	speed-proportional scale factor
<code>bias_omega</code>	per-wheel $\mathcal{N}(0, 0.2^2)$ rad/s	drawn once per run — a <i>bias</i> , absorbed by the slip states, never folded into R

Table 5: Optical flow, 100 Hz (`FlowModel`).

Field	Value	Provenance note
<code>sigma_flow0</code>	0.005 m/s	white-noise floor (near-zero-speed regime)
<code>sigma_flow</code>	0.02 (2% of speed)	17.6–25.7 mm std over 1 m trials (PMC4481980)
<code>sf_flow</code>	0.147	height-scale error, 14.7%/mm, drawn <i>once per run</i> — excluded from R
<code>quantum</code>	$31.75 \mu\text{m}/\text{count}$	ADNS-3080 @ 800 cpi
<code>v_max_track</code>	1.02 m/s	loss of lock above this $\Rightarrow$ <i>no</i> measurement
<code>dropout_prob</code>	0.02	per-sample dropout (low-texture floor)

D The tuning pipelines, with algebra

D.1 Common optimizer protocol (all tunings)

Two-phase staged search (`controller_tuning/optimizer_stage.jl`):

1. **Global:** **dxNES** (BlackBoxOptim) on a cheap screen objective, per-seed Sobol-offset start, run in chunks with a *plateau gate* (relative improvement below `rel_tol` over a trailing window) and a hard eval cap (`--p1-cap 250`). Convergence is detected, never assumed — the v1 tuner was caught terminating mid-descent.
2. **Local:** **BOBYQA** (NLOpt LN_BOBYQA) on the full objective inside a box centred on the phase-1 optimum (`--p2-cap 60`), with an explicit score-plateau check — BOBYQA’s own `xtol/ftol` criterion was measured to *not* fire even when the score had plateaued.

Controllers: 5 seeds clean (cold start), 4 seeds noisy (warm-start from the same run’s clean optimum). Estimator: 5 seeds, replay. One OS process per seed; BLAS single-threaded; `stop_reason` recorded per run.

D.2 Controller objective (`stage_objective.jl`)

$$\text{score} = \text{tracking}_{v3} + 0.05 \frac{ce}{24} + 0.36 \frac{\text{chatter}}{0.8}, \quad (31)$$

with the six-term tracking (errors in physical units, tolerances normalising each term; $he = 2|\sin(\Delta\psi/2)|$, $pe = \sqrt{\Delta x^2 + \Delta y^2}$):

$$\text{tracking}_{v3} = \frac{1}{6 k_{traj}} \left(\frac{pe_T}{0.01} + \frac{\max_t pe}{0.1} + \frac{\overline{pe}}{0.0249} + \frac{he_T}{0.01} + \frac{\max_t he}{0.1} + \frac{\overline{he}}{0.0128} \right), \quad (32)$$

$$k_{traj} = \frac{\max(\max_t \|p_{ref}(t) - \bar{p}_{ref}\|, 0.1)}{0.1} \geq 1 \quad (\text{path extent, rotation-invariant}), \quad (33)$$

Table 6: Pose fix, :docking tier, 100 Hz (PoseFixModel).

Field	Value	Note
sigma_pos	0.01 m	absolute (x, y) fix
sigma_psi	0.5°	heading fix
outlier_frac	0.005	gated at 0.5 m
dropout_frac	0.0	no dropouts in this tier

Table 7: Noisy-oracle injection (OracleEstimator(:noisy), hybrid_ctrl/estimators.jl; scale = 1.0). Model: meas = true + bias + $(\sigma_0 + SF \cdot |\text{true}|)\mathcal{N}(0, 1)$, biases drawn once per run. This is the *pessimistic no-filtering bound* of §2 — unfiltered noise written straight into the controller’s state estimate, no estimator in the path.

Channel	σ_0	SF	turn-on bias std
velocity V_x, V_y	0.010 m/s	0.02	0.005 m/s
yaw rate $\dot{\psi}$	0.003 rad/s	0.005	0.003 rad/s
heading ψ	0.010 rad	—	0.005 rad
position X_o, Y_o	0.020 m	—	0.010 m

terminal / peak / typical per channel, $\overline{(\cdot)}$ the time-normalised mean absolute. The 0.0249 / 0.0128 tolerances are *measured* (achieved iae/max ratios 0.249 / 0.128 over 7 converged configs $\times$ 14 trajectories). $ce = \frac{1}{T} \sum_t \sum_{i=1}^4 |v_{cmd,i}|$ (voltage magnitude); $chatter = \frac{1}{T} \sum_t \sum_i |\Delta v_{cmd,i}|$ per ms tick (slew rate), referenced to the mixer’s own clamp $4dV_{max}/f_{mix} = 0.8$ V/ms.

D.3 PID-FB / PID-CT — IMC cascade, 3 tuned dims

Per-axis inner plant $m_{eff}\dot{v} + d_{eff}v = W$ with $m_{eff} = (R\tilde{m}, R\tilde{m}, I_\psi) \approx (2.25, 2.25, 5.89)$ and d_{eff} the R -scaled rows of D_{eq} . IMC pole cancellation:

$$K_p = \frac{m_{eff}}{\lambda}, \quad K_i = \frac{d_{eff}}{\lambda}, \quad K_d = 0 \text{ (a result, not a choice),} \quad (34)$$

$$\frac{K_i}{K_p} = \frac{1}{\tau_{open}} \text{ identically,} \quad \tau_{open} = \frac{m_{eff}}{d_{eff}} \approx (0.256, 0.114, 0.155) \text{ s.} \quad (35)$$

Outer loop on the integrator plant $\dot{p} = v$: $K_{p,pos} = 1/(N\lambda)$, $N = 4$, giving $\omega_n = 1/(\lambda\sqrt{N})$, $\zeta = \sqrt{N}/2 = 1$ (critically damped). Anti-windup is *derived*: $I_{max} = \text{wrench_budget}/K_i$ with the budget the measured p95 integral demand, so $K_i I_{max}$ is invariant of λ . PID-CT adds the computed-torque feedforward

$$W_{ff} = m_{eff} a_{cmd} + d_{eff} V_{cmd} + C(\dot{\psi})\hat{V}, \quad (36)$$

structurally matched to ASMC’s equivalent control. Tuned space (PID_SPACE_V2, log): $\lambda_x \in [0.003, 0.256]$, $\lambda_y \in [0.003, 0.114]$, $\lambda_\psi \in [0.003, 0.155]$ — upper bounds *are* τ_{open} (beyond which the loop is slower than open loop); the lower bound is a numerical guard at $3dt$, the real floor being priced endogenously by the chatter term ($K_p = m_{eff}/\lambda$ amplifies command noise as $\lambda \rightarrow 0$).

D.4 ASMC v2 — adaptive supertwist, 4 tuned dims

Surface per axis with a saturation-aware slope, $\lambda(e) = \min(\lambda_{max}, v_{max}/|e|)$; smooth switching wrench $W = W_{eq} + K \tanh(s/\varepsilon)$ with the boundary layer *derived* from estimator noise, $\varepsilon_i =$

$\max(0.5 \text{tol}_i, k \sigma_{s,i})$. Gain adaptation (per axis) with the friction-circle ceiling recomputed per tick and the demand-driven decay:

$$\dot{K}_i = \gamma_i \left[s_i \tanh(s_i/\varepsilon_i) (1 - z_i) - \rho_{auth} \varepsilon_i z_i \right], \quad z_i = \frac{K_i - K_{floor}}{K_{max,sched,i} - K_{floor}}, \quad (37)$$

so K grows under sliding-mode activity, parks below the scheduled ceiling $K_{max,sched}$ (the wheel-3 friction budget, `kmax_schedule`), and decays toward the floor at a rate set by the demand scale $\rho_{auth}\varepsilon$; γ is a pure time-scaling (fixed at the derived per-axis values 170/380/280 after a $64\times$ sweep showed it score-inert). Tuned space (`ASMC_SPACE_V2`, log): $\lambda_{x,max}, \lambda_{y,max} \in [0.3, 10]$, $\lambda_{\psi,max} \in [1, 20]$, $\rho_{auth} \in [0.762, 27]$ (both ends physically anchored: $\tanh(1)$ and the σ -leak excitation scale).

D.5 Estimator tuning — replay objective, 7 tuned dims

The estimator is tuned *open loop* (replay against ground truth on logged trajectories; no controller in the loop), same tolerance-normalised shape:

$$\text{score}_{est} = \frac{vel_{rmse}}{10^{-3}} + \frac{rate_{rmse}}{10^{-2}} + 1.0 \cdot \frac{inslip_vel_{rmse}}{10^{-3}} + \frac{pos_{rmse}}{10^{-2}} + \frac{head_{rmse}}{10^{-2}} + 0.05 \cdot \text{smoothness}, \quad (38)$$

velocity units m/s, rates rad/s, positions m, headings rad; smoothness = $\text{mean}((d|\hat{v}|/dt)^2)$. The in-slip velocity error is priced *twice* (once in vel_{rmse} , once gated) — the estimator’s model mismatch on this platform *is* slip. Search space (`param_space_v4.jl`, all log-scaled): $\alpha_{acc}, \alpha_{yaw} \in [10^{-2}, 10^2]$, `grip_slip_scale` $\in [10^{-5}, 1]$, `r_boost` $\in [1, 100]$, `pose_Qn_heading` $\in [10^{-8}, 10^{-3}]$, `slip_R_inflate` $\in [1, 100]$, $q_\alpha \in [10^{-8}, 10^{-1}]$. Conditions: `train12`, $\mu = 0.5/\chi = 0.005$, `:realistic` suite, $f_{est} = 1$ kHz, `fix_tier=:docking`. All five seeds converge to the same estimator within 0.23% on a common noise realisation; seed 4 adopted and frozen.